

Module Manual V1.0

Pinout:

Pin1	VCC	Power + (3v-5v)
Pin 2	RX	Receiving data from serial
Pin 3	TX	Serial data transmission
Pin 4	GND	Power Ground
Pin 5	RST	Internal use, no connection, vacant
Pin 6	B0	Internal use, no connection, vacant
Pin 7	SCL	I2C clock
Pin 8	SDA	I2C data

Facts:

Measuring range	-180 ° 1 °
Resolution	0.01 °
measurement accuracy	1
Repeatability	1
Frequency response	100 HZ (115200bps)
Operating Voltage	35 V
Working current	15mA
Operating temperature	-20 ° 8 °
Storage temperature	-40 ° 1 °
size	11.5mm × 15.5mm

Communication protocol:

UART

Baud rate: 115200 bps	Parity bit: N	Data bits: 8	Stop bits: 1 (DEFAULT)
Baud rate: 9600 bps	Parity bit: N	Data bits: 8	Stop bits: 1

Note: The baud rate selection is operated by solder jumper, see below.

Output format :

each frame contains 8 bytes (hex):

- | | | |
|------------|-----------|-------------------|
| (1).Byte0: | 0xAA | Preamble Flags |
| (2).Byte1: | 0x00-0xFF | HIGH heading high |
| (3).Byte2: | 0x00-0xFF | LOW heading lower |
| (4).Byte3: | 0x00-0xFF | HIGH pitch angle |
| (5).Byte4: | 0x00-0xFF | LOW pitch angle |
| (6).Byte5: | 0x00-0xFF | HIGH roll angle |
| (7).Byte6: | 0x00-0xFF | LOW roll angle |
| (8).Byte7: | 0x55 | Frame end flag |

Calculation method:

Angle = ((HIGH << 8) | LOW) / 100;

Example:

Given the following data [0xAA, 0x00, 0x64, 0x03, 0xE8, 0x27, 0x10, 0x55]

heading angle = 1.00 °

Pitch angle = 10.00 °

Roll Angle = 100.00 °

COMMANDS:

1. 0xA5 + 0x51: query mode, return directly to the angle value, to be sent each read
2. 0xA5 + 0x52: Automatic mode, send a direct return angle, only initialization
3. 0xA5 + 0x53: Automatic mode, ASCII code output, serial port for direct computer assistant View
4. 0xA5 + 0x54: correction mode, the pitch correction roll angle of 0 degrees, need to stay level when sending
5. 0xA5 + 0x55: correction mode, 0 degree course correction, heading cleared at any angle

(1), due to self-correction at power up the module must be held in a stationary position for at least 3000ms, hand-held of this module is not recommended

(2), module heading will drift after a while without magnetometer.

(3), due to the angle of the Euler angles universal lock problem, roll, pitch, have an impact on each other at 90 degrees.

(4), The module IOs are 5.0V tolerant, the module can be used with 5.0 and 3.3V system and serial adapters without any risk.